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In [1]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt
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In [2]: # Create and connect to SQLite database
conn = sqlite3.connect("sales_data.db")
cursor = conn.cursor()
```

```
In [3]: # Create sales table
cursor.execute("DROP TABLE IF EXISTS sales")
cursor.execute("""
    CREATE TABLE sales (
        sale_id INTEGER PRIMARY KEY,
        product TEXT,
        quantity INTEGER,
        price REAL
    )
""")
```

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Out[3]: <sqlite3.Cursor at 0x15ba37082c0>
```

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In [4]: # Insert sample data
products = ['Laptop', 'Monitor', 'Keyboard', 'Mouse', 'Headphones']
import random
data = [(random.choice(products), random.randint(1, 10), round(random.uniform(50, 1000), 2)) for _ in range(30)]

cursor.executemany("INSERT INTO sales (product, quantity, price) VALUES (?, ?, ?)", data)
conn.commit()
```

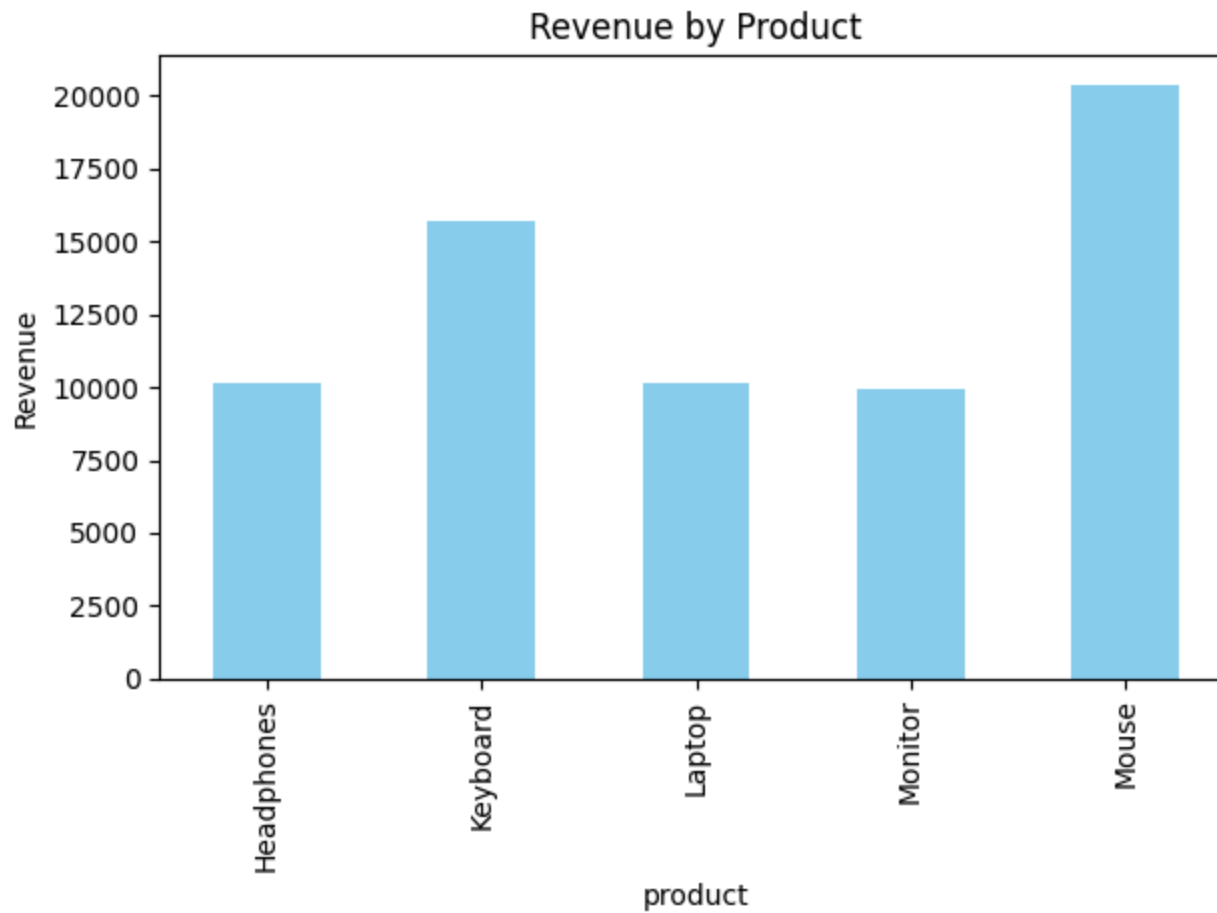
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In [5]: # Query sales summary
query = """
    SELECT product,
           SUM(quantity) AS total_qty,
           SUM(quantity * price) AS revenue
    FROM sales
    GROUP BY product
    """
```

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In [6]: df = pd.read_sql_query(query, conn)
conn.close()
```

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In [7]: # Print result
print(df)
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	product	total_qty	revenue
0	Headphones	16	10124.02
1	Keyboard	47	15665.36
2	Laptop	21	10155.68
3	Monitor	24	9918.04
4	Mouse	35	20349.61

```
In [8]: # Plot bar chart
df.plot(kind='bar', x='product', y='revenue', legend=False, color='skyblue')
plt.title("Revenue by Product")
plt.ylabel("Revenue")
plt.tight_layout()
plt.savefig("sales_chart.png")
```



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In [ ]:
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